



EUROPEAN CENTRE FOR MEDIUM-RANGE WEATHER FORECASTS

Scientist for Machine Learning

1. Position information

Vacancy No.: VN20-15	Department: Research
Grade: A2	Section: Model Development
Job Ref. No.: STF-PL/20-15	Reports to: Physical Processes Team Leader
Publication Date: 25 September 2020	Closing Date: 22 October 2020

2. About ECMWF

ECMWF is the leading centre for global, medium-range weather predictions and is the host of the largest archive of weather data in the world. ECMWF is both a research institute and a 24/7 operational service, producing and disseminating numerical weather predictions to its Member States.

ECMWF has been also entrusted to operate the Copernicus Atmosphere Monitoring Service (CAMS) and the Copernicus Climate Change Service (C3S) on behalf of the European Commission. Every day, hundreds of millions of satellite and in situ observations are processed at ECMWF to provide the basis for up-to-date global analyses and climate reanalyses of the atmosphere, ocean and land surface, and to generate global weather predictions from hours up to a year ahead. To retain its world-leading position, ECMWF is performing cutting edge research in weather related sciences and high-performance computing. ECMWF's weather forecasts are disseminated to the ECMWF Member States and thousands of users around the world.

For details visit www.ecmwf.int/.

3. Summary of the role

ECMWF has embarked on an exciting new initiative to explore the use of artificial intelligence and machine learning in applications of numerical weather predictions. To learn more about the application of machine learning in the weather and climate domain and at ECMWF, please have a look at the webpage of the Machine Learning Seminar Series at ECMWF (<https://www.ecmwf.int/en/learning/workshops/machine-learning-seminar-series>) or the ESA-ECMWF machine learning workshop which is planned for October (<https://www.ecmwf.int/en/learning/workshops/ecmwf-esa-workshop-machine-learning-earth-system-observation-and-prediction>).

As part of this effort, ECMWF is coordinating the MACHiNE Learning for Scalable meTeoROlogy and climate (MAELSTROM) EuroHPC project to fund this position. This Scientist position will be in the Physical Processes team in the Research department at ECMWF. The successful candidate will apply their skills, knowledge and expertise to help achieving the goals, and complete the deliverables, of the MAELSTROM project.

The main focus will be on the development of machine learning emulators for some of the parameterisation schemes of ECMWF's Integrated Forecast System (IFS). The use of deep learning to emulate some of the parametrizations used to represent subgrid atmospheric processes, such as radiation or clouds, promises a significant reduction of computing cost and improvements in portability of the models to heterogeneous hardware, and could potentially lead to improvements in predictive skill if savings are reinvested into higher resolution or model complexity. This project builds on previous studies which have successfully used neural network emulators within weather and climate simulations. However, the use in a forecasting system for operational weather predictions, such as the IFS, will require more complex machine learning solutions and training data of higher quality than what has been used so far. The successful candidate will also explore the use of such emulators within the data assimilation framework.

The Scientist will work in close collaboration with other teams across the organisation and strong communication skills are essential.

4. Main duties and key responsibilities:

- Diagnosing training data in form of input/output pairs of physical parametrisation schemes from simulations with the IFS at high resolution
- Publishing the training data in a user-friendly form for use as benchmark dataset for MAELSTROM
- Developing customised machine learning solutions for the emulation of parametrisation schemes
- Reintroducing machine learning solutions into the IFS source code and evaluate model fidelity in coupled forecast and data assimilation experiments
- Testing machine learning emulators for data assimilation applications
- Contributing to reports, and dissemination and training activities of the MAELSTROM project

5. Personal attributes

- Strong interpersonal and communication skills, particularly listening to and respecting the views of others
- Enthusiasm to tackle challenging research questions when working with a complex computer model and willingness to learn new algorithms and tools
- Ability to work in a team at ECMWF and within MAELSTROM towards a common goal in an interdisciplinary research project
- Excellent analytical and problem-solving skills with an independent and proactive approach, together with an interest in identifying, investigating and solving technical challenges

6. Qualifications and experience required

Education	A university degree, or equivalent, in a discipline related to computer science, meteorology, physics, mathematics, machine learning or engineering is required. A PhD in a related subject is desirable but not essential.
Experience	Experience in developing codes in parallel computing environments. Experience working with numerical models in computational fluid dynamics.

	Experience working with deep-learning and neural networks would be advantageous. Experience working with global weather or climate models would be advantageous. Experience with using Python for meteorological data, in particular machine libraries such as TensorFlow or Keras, would be advantageous.
Knowledge and skills (including language)	Good knowledge of at least one high-level programming language such as C++ or Fortran. Ability to work in a Linux-based environment. A good knowledge of Python and Jupiter notebooks would be useful. Good understanding of parallel programming (e.g. MPI and OpenMP) would be useful. Candidates must be able to work effectively in English and interviews will be conducted in English. A good knowledge of one of the Centre's other working languages (French or German) would be an advantage.

7. Other information

Grade remuneration

The successful candidate will be recruited at the **A2** grade, according to the scales of the Co-ordinated Organisations and the annual basic salary will be **£60,590.64** net of tax. This position is assigned to the employment category **STF-PL** as defined in the Staff Regulations.

Full details of salary scales and allowances are available on the ECMWF website at www.ecmwf.int/en/about/jobs, including the Centre's Staff Regulations regarding the terms and conditions of employment.

Starting date: 1st January 2021, or as soon as possible thereafter.

Length of contract: 30 months, subject to available funding with possibility of extension.

Location: The position will be based in the Reading area, in Berkshire, United Kingdom.

Successful applicants and members of their family forming part of their households will be exempt from immigration restrictions.

Videoconference interviews (via Blue Jeans) for this position are expected to take place in mid-November.

8. How to apply

Please complete the online application form available at: www.ecmwf.int/en/about/jobs.

To contact the ECMWF Recruitment Team, please email jobs@ecmwf.int.

Please refer to the ECMWF Privacy Statement. For details of how we will handle your personal data for this purpose, see: <https://www.ecmwf.int/en/privacy>.

At ECMWF, we consider an inclusive environment as key for our success. We are dedicated to ensuring a workplace that embraces diversity and provides equal opportunities for all, without distinction as to race, gender, age, marital status, social status, disability, sexual orientation, religion,

personality, ethnicity and culture. We value the benefits derived from a diverse workforce and are committed to having staff that reflect the diversity of the countries that are part of our community, in an environment that nurtures equality and inclusion.

Staff are usually recruited from among nationals of the following Member States and Co-operating States:

Austria, Belgium, Bulgaria, Croatia, Czech Republic, Denmark, Estonia, Finland France, Hungary, Germany, Greece, Iceland, Ireland, Israel, Italy, Latvia, Lithuania, Luxembourg, Montenegro, Morocco, the Netherlands, North Macedonia, Norway, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.
Staff from non-ECMWF States may be considered in exceptional cases.